

Catch every strip defect where it is born, not where it is shipped

AI surface inspection that finds, grades and routes every stainless strip defect to the process that caused it.

90%

of defects caught on real mill strip it had never seen

4.7%

of clean strip wrongly flagged, fewer than 1 in 20

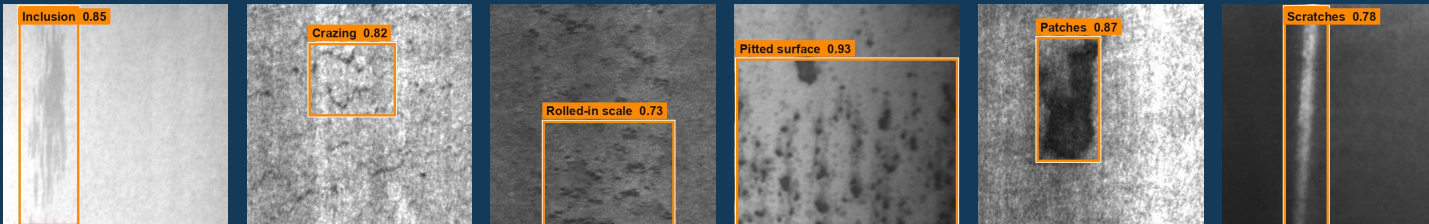
₹2.5-4.2 cr

per line, vs ₹10-29 cr for a commercial system

~2.2 years

payback on one line at a 1% downgrade rate

REAL DETECTIONS BY OUR MODEL, ON STEEL IMAGES IT NEVER SAW IN TRAINING



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TRY THE LIVE DEMO

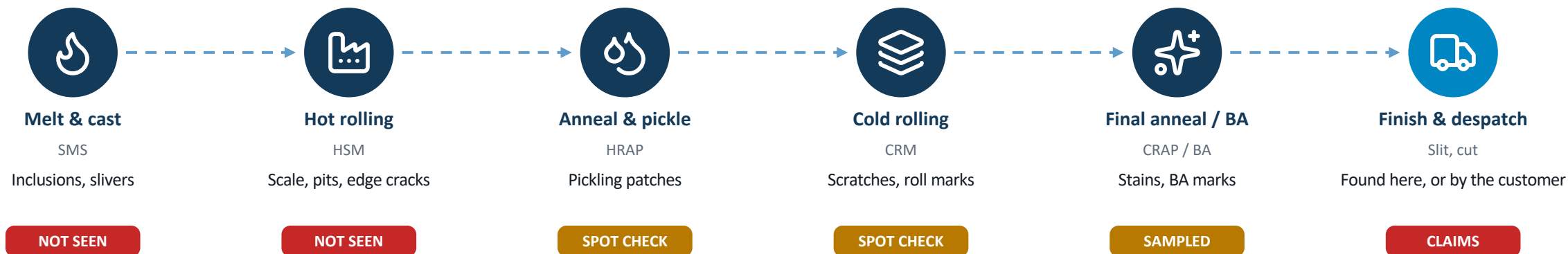
surface-vision.github.io

Runs in your browser. Nothing leaves the page.

A defect born at the caster is paid for at every later stage, but today it is found only at the end

Where stainless surface defects are born, and where they are seen today

VALUE ADDED KEEPS RISING →

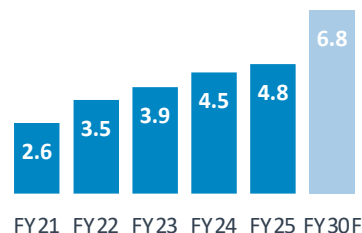


Why it hurts more on stainless

- 300 m/min**
too fast for the eye to see the whole surface
- ₹126k-165k**
of nickel in every tonne of 304, given away on a downgrade
- No repair**
damage on 2B/BA mill finishes cannot be polished out

Why now

India demand, Mt +84% in 5 yrs



- 1.73 Mt imports**
FY25; anti-dumping probe on cold-rolled 300/400, Sep 2025
- IS 6911 mandatory**
BIS certification for sheet and strip, since Aug 2024
- 2.67 MTPA**
JSL cold rolling by FY28, up from 2.05

Cost of quality: where AI moves the money

← spend moves from failure to prevention

Prevention defect sent to its owner in minutes	Internal failure caught before more value is added
Appraisal 100% of the surface, every coil	External failure fewer escapes to customers

Poor quality costs ~15% of sales in manufacturing (IISE).

Sources: AMETEK/Ternium (2020); ASSDA; IISE; ASTM A240; LME nickel USD 16,402/t at INR 95.8/USD (22 Sep 2026); ISSDA (FY25); DGTR (29 Sep 2025); BIS Steel QCO 2024; JSL.

The problem is not that defects exist. They are found late, by sampling, and nobody is told which process caused them.

On stainless the surface is the product, so every defect type must reach the process that caused it

Six defect families, six different owners (real detections by our model)

Born: **Caster**
Owner: **SMS**
Fix: Tundish level, mould flux

Born: **Reheat, hot rolling**
Owner: **HSM**
Fix: Reheat and coiling temperature

Born: **Hot rolling**
Owner: **HSM**
Fix: Descaler pressure, nozzles

Born: **Reheat, pickling**
Owner: **HSM / pickling**
Fix: Furnace atmosphere, over-pickling

Born: **Anneal & pickle**
Owner: **Pickling line**
Fix: Acid strength, line speed

Born: **Cold rolling, handling**
Owner: **CRM / logistics**
Fix: Guides, rolls, coil handling

Stainless-only types to add with JSL data:

Roping

Ridging (ferritic 430)

Orange peel

Anneal colour

BA stains

Lamination

Precedent: who already does this, and what we copy

Outokumpu	Europe's #1 stainless maker; AI inspection live at Tornio since 2024, runs offline. Copy: offline edge, multi-site
Tata Steel	Kalinganagar (WEF Lighthouse) finds cold-rolled surface defects by video analytics. Copy: in-house model team
Parsytec	On stainless lines: 170 µm cameras, bright + dark-field light, a decision per coil. Copy: optics, coil decision

Grade-by-defect risk: which defects to train for first

Grade	Slivers	Scale	Scratches	Edge cracks	Ridging
200 series (Cr-Mn)	M	M	H	M	L
300 series (304, 316)	H	M	H	L	L
400 series (409, 430)	H	M	M	M	H
Duplex (2205)	M	H	M	H	L

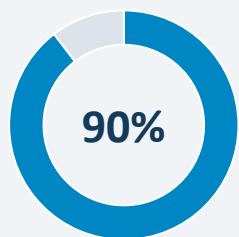
OUTPUT
300 series is the pilot grade: highest exposed-finish risk and the most nickel per tonne.

Images: held-out test frames, boxes by our model. Sources: Nortal/Outokumpu (2024); Tata Steel; Parsytec (Stainless Steel World); Leão 2021; Steel in Translation 2023. Owners: team hypothesis. Grade risks: team hypothesis.

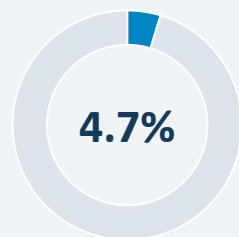
Classify, don't just alarm: a typed defect becomes a work order for a named owner. The pilot belongs on 300-series cold-rolled strip.

Four design choices behind the system, each proven in our own tests

1 Trained on clean steel too, so its alarms mean something



of defects caught
258 of 288 defective mill
images



of clean strip flagged
14 of 300 clean mill
images

So we: train on JSL's own clean and defective strip before any alarm is trusted.

2 A compact model is enough

12 MB

the whole model; loads in any
browser

~55 ms

per image on an ordinary laptop
CPU

0.75

accuracy, inside the 0.70-0.80
published range

Larger models and sharper images added no accuracy in our tests.

So we: keep the model compact and spend the budget on camera coverage.

3 Image detail, not the model, sets the hardware bill

0.2 mm per pixel



~18 processors

sees defects from ~0.5 mm

0.5 mm per pixel



~3 processors

sees defects from ~1.3 mm

1.28 mm per pixel



1 processor

fine scratches vanish (~3 mm)

1.28 m strip at 250 m/min. Processors scale with pixels; a defect needs 2-3 pixels.

So we: fix the image resolution with JSL first, then size the hardware.

4 Operators act only on alarms they can trust



±3 pts

between stated confidence and reality, so a "70%" alarm is right
about 70% of the time



0 of 3,200

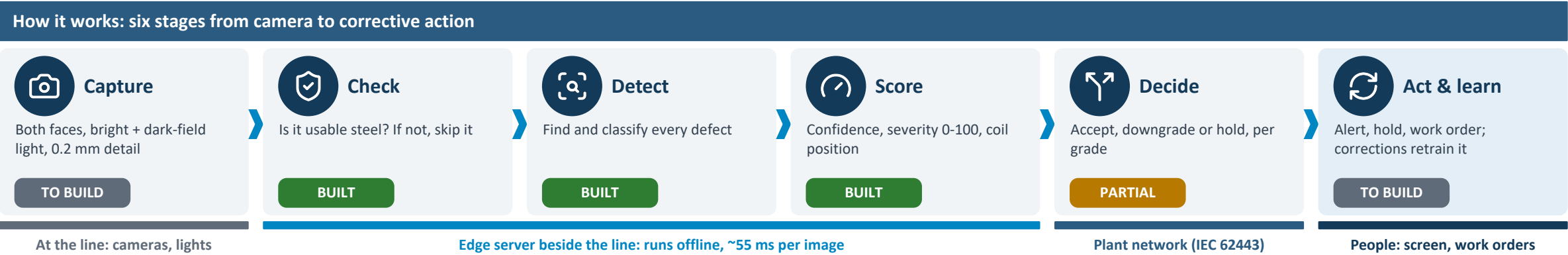
real steel images wrongly rejected by our "is this steel?" check,
which stops the system judging anything else

So we: every alarm carries a trustworthy confidence and a steel check.

All numbers measured by us on three public steel-surface datasets, on images the model never saw in training.

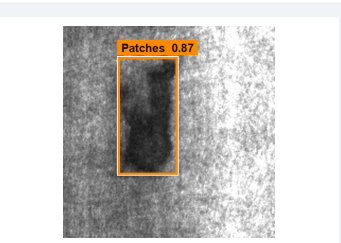
The model is the cheap part. JSL's own strip, the image resolution and trustworthy confidence decide whether the system works.

Detect on the strip, decide at the coil, route to the owner, and learn from every correction



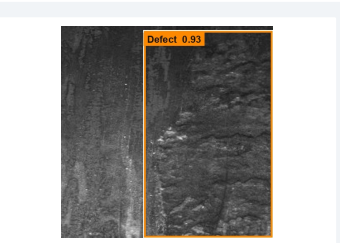
It already works on public steel data; the pilot's job is to prove it on JSL's own strip

Tested on three public steel datasets



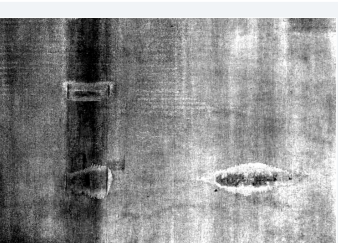
NEU-DET
1,800 images, 6 types

University benchmark, hot-rolled strip



Severstal
12,568 real mill frames

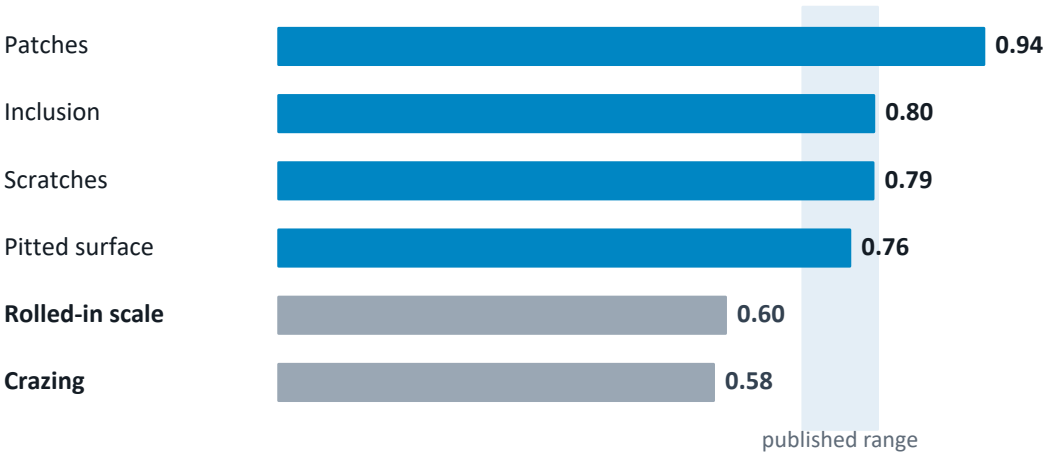
Production strip from a steel mill



GC10-DET
2,294 frames, 10 types

Full-width line-scan camera frames (roll marks shown)

How well it finds each defect type (accuracy, held-out images)

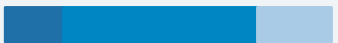


Headline results


178 of 180
test images: the right defect type named


~55 ms
per image, running in an ordinary laptop browser


~10,100



■ Lab 1,800
■ Mill strip 6,001
■ Line-scan 2,294

Known weak spots, and the fix for each

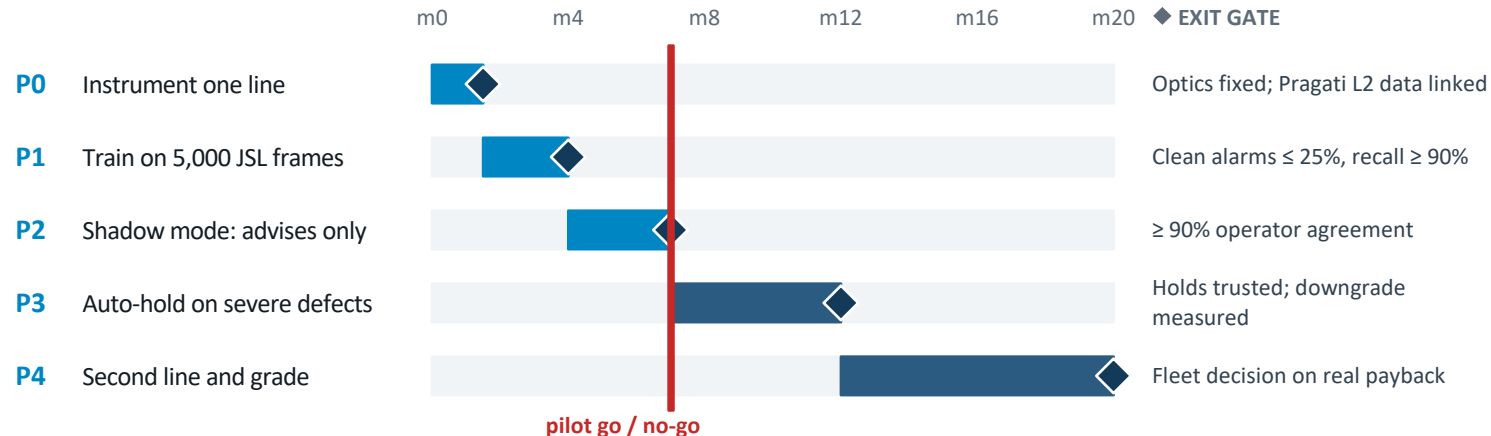
Weak spot	Today (0-1)	Fix in the pilot
Low-contrast texture: crazing, rolled-in scale	0.58-0.60	Angled lighting; JSL labels in P1
Real mill frames are harder than the lab benchmark	0.57 vs 0.75	Train on JSL's own strip in P1
Roll marks and edge cracks: few examples	0.20	Edge cameras; synthetic defects; targeted labels

Datasets: NEU-DET (Northeastern Univ.); Severstal steel defects (Kaggle); GC10-DET (Lv et al. 2020, CC BY 4.0). All scores on held-out images the model never saw in training; model chosen on validation across four training runs.

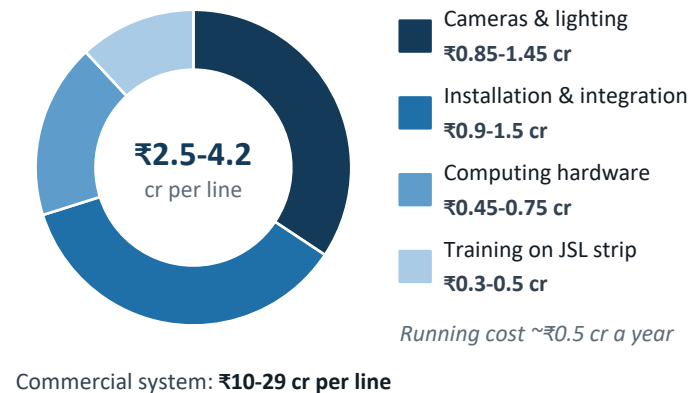
Strong on four of six defect families today. Every weak spot is known, and each has a fix in the pilot plan.

Seven months to shadow mode on one line, and nothing touches the mill until trust is earned

Stage-gate roadmap (months)



Cost per line, estimated



Top risks, and when each is retired

Risk	Mitigation	Retired
Model does not transfer to JSL strip	Train on JSL clean and defect images	P1
Mirror-like BA/2B finish hides defects	Bright + dark-field light; deflectometry for BA	P0-P1
Rare defects under-learnt	Edge cameras; synthetic defects; targeted labels	P1-P3
Operators ignore alarms	Trusted scores; shadow mode first	P2
Plant network, heat and fumes	IEC 62443 zones, runs offline; sealed enclosures	P0
Quality slips after a model update	Every retrain must beat the live model first	P2+

Pilot KPIs: how JSL will know it works

$\geq 90\%$

defective strip flagged
measured 258 of 288 (89.6%)

$\leq 25\%$

clean strip wrongly flagged
measured 4.7% on mill strip

≤ 5 pts

confidence vs reality gap
measured 3.3 pts

< 0.2 s

defect to operator screen
design target

$\geq 90\%$

operator agreement, shadow
measured in P2

< 8 h

retrain for a new grade
measured 0.6 h

Costs: team estimates for a 1.3-1.6 m line, both faces, at INR 95.8/USD; vendor quotes replace them in P0. Pragati: JSL digitalisation, Phase 2 brings Level-2 data at Jajpur.

Commit ₹2.5-4.2 cr and one line for seven months. The month-7 go/no-go is decided on JSL's own alarm and recall numbers.

At a conservative 1% downgrade rate one line pays back in ~2.2 years; the fleet protects ~₹13 cr a year

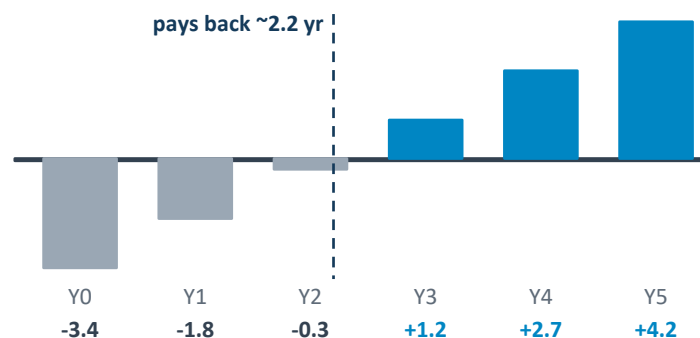
Loss pool per line, ₹ cr a year

$0.4 \text{ Mt} \times \text{downgrade rate} \times \text{price discount} \times ₹167,407/\text{t}$

Downgraded	10% off	15% off	20% off
0.5% of volume	3.3	5.0	6.7
1.0% of volume	6.7	10.0	13.4
2.0% of volume	13.4	20.1	26.8

Base case (dark): ₹6.7 cr pool, ₹2.0 cr/yr recovered (₹1.5 cr net). Fleet: ₹12.9 cr/yr. Non-prime coil sells 10-30% below prime, so these discounts are conservative.

Cumulative cash per line, ₹ cr

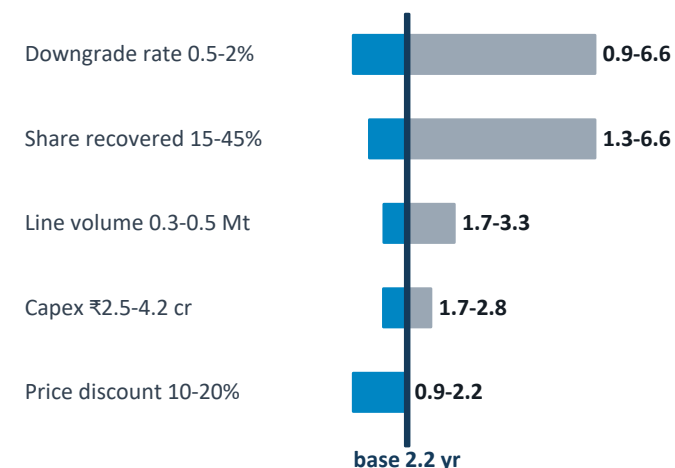


+₹2.1 cr 5-year NPV per line at 12%

-₹11.6 cr commercial system after 5 years

≥ 0.11% of revenue lost to downgrade = 2-yr payback

What moves payback (years)



Impact beyond the P&L



Coverage

100% of both faces, every coil

today: spot checks



Customers

Defect map per coil for rail, metro, auto buyers

Vande Bharat supplier



Process

Defects tied to heat and coil IDs

6 families, 6 owners



Safety

Severe-defect holds prevent strip breaks

Ternium's business case



Sustainability

Less rework and remelt; backs net zero

JSL: 1.76 tCO2e/t, FY26



People

Inspectors move to decisions

HR AI: 95% self-served

WHAT WE ASK OF JSL

- One 300-series cold-rolled line, 7 months
- ₹2.5-4.2 cr pilot budget
- Downgrade rate by line and grade
- A metallurgist at 20% for root causes

In return: measured downgrade rate, payback and a go / no-go at month 7.

Inputs: JSL FY26 (INR 167,407/t; 2,565,902 t). Base: 0.4 Mt line, 30% of loss recovered, ₹3.35 cr capex, ₹0.5 cr/yr run cost, 12% rate. Commercial: USD 2M at INR 95.8.

Recommendation: approve a 7-month pilot on one 300-series cold-rolled line. JSL's own downgrade rate decides fleet rollout.